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Retrieval Strategies for Scientific Equipment Discovery: From BM25 to Weighted Hybrid Fusion

Bachelor's Project Thesis

Viktor Kotuliak, s5617952, v.kotuliak@student.rug.nl
Supervisor: prof J.S. Dibangoye

Abstract: Keyword-based search over large equipment databases often struggles when user queries rely on natural language rather than exact terminology. This paper presents a progressive ablation study evaluating retrieval systems for equipment discovery, comparing the sparse encoder BM25, two dense encoders (`all-mpnet` and E5), and a hybrid fusion of BM25 and E5 using Weighted Reciprocal Rank Fusion. Retrieval systems are evaluated on a synthetically generated benchmark of query-document pairs, enabling controlled comparison across retrieval strategies. The hybrid architecture achieved the best performance, reaching an nDCG@10 score of 0.6845 and outperforming both dense and sparse retrievers in isolation. These results suggest that combining lexical and semantic signals through rank fusion offers an effective strategy for retrieval in domains with specialised vocabulary, within the scope of this controlled benchmark; validation against real user queries remains an important direction for future work.

1 Introduction

As university research infrastructure expands, the equipment catalogued in institutional repositories grows in both volume and diversity, making it harder to discover efficiently. At the University of Groningen, the database currently indexes over 8,000 instruments; however, its keyword-based search interface requires researchers to know the precise name or category of the equipment they seek. When a researcher can only describe an experimental need, rather than a specific instrument, the existing system fails to surface relevant results, directly hindering experimental efficiency.

To address this, the thesis proposes a hybrid retrieval system that serves as the foundational retrieval component of a Retrieval-Augmented Generation (RAG) pipeline. RAG augments a language model's generative capacity with a retrieval component that queries an external knowledge base at inference time [Gao et al., 2023, Lewis et al., 2020]. This approach is particularly suited to the domain because the equipment catalogue is updated continuously, and retrieval-based systems can incorporate new documents without retraining, a critical advantage over fine-tuning-based alternatives. Furthermore, RAG confines responses to grounded, retrievable evidence, reducing hallucination risk and ensuring factual accuracy. Rather than requiring researchers to memorise equipment names, the proposed system allows them to describe their experimental context in natural language; the retrieval component then identifies and ranks the most appropriate instruments, together with metadata such as short description and research-oriented use.

The scope of this thesis is deliberately bounded: it focuses on the retrieval component of the RAG pipeline, systematically comparing and evaluating multiple retrieval strategies: sparse (BM25), dense (bi-encoder), and hybrid, rather than examining the downstream generation stage. These considerations led to the following research question: **How do different retrieval strategies and embedding techniques impact the accuracy of equipment recommendations in a RAG-based search system?**

2 Related Work

This section surveys the literature underpinning RAG systems, covering data preparation, retrieval, generation, and evaluation. Figure 2.1 provides a conceptual overview of how these stages interact within a unified framework.

2.1 Data Preparation

Following Gao et al. [2023], the data preparation stage comprises three phases: **data indexing**, **chunking** and **embedding**

2.1.1 Data indexing

Cleaning and extracting the data from different file formats and converting them to plain text.[Gao et al., 2023].

2.1.2 Chunking

Chunking addresses a core limitation of LLMs: their finite context windows prevent processing of long documents in a single pass, requiring their division into smaller segments, generally through rule-based fixed-length segmentation [Wang et al., 2025].

Chunk size involves an inherent trade-off: larger chunks preserve more context but may introduce noise, and are further disadvantaged by the tendency of LLMs to attend more reliably to information appearing early in the input than to information appearing later [Gao et al., 2024]. Smaller chunks, meanwhile, risk fragmenting the information needed for coherent reasoning. However, the practical impact of chunk size on end-to-end system performance is not always pronounced; Li et al. [2025] find only marginal differences across chunk sizes in their RAG benchmark, suggesting that other pipeline components may have a stronger influence on output quality. Nonetheless, chunking strategy remains a relevant design consideration, particularly when retrieval rather than generation quality is the primary objective.

2.1.3 Embedding

Embeddings are vector representations of text that project natural language into a continuous, high-dimensional space, enabling systems to operate on language in a mathematically tractable format. In a RAG pipeline, both the knowledge base documents and incoming user queries are encoded into this shared space, so that retrieval reduces to a geometric proximity problem.

Semantic representation. Unlike sparse lexical models such as `BM25`, dense embeddings determine similarity by meaning rather than surface-form overlap, mitigating the lexical mismatch problem.

Fine-tuning embeddings. Pre-trained embedding models are trained on large general-domain corpora and may therefore fail to capture the terminology and query patterns specific to a target domain [Wang et al., 2022]. Fine-tuning adapts a pre-trained encoder to a retrieval task by continuing training on domain-relevant query-document pairs. A common approach is contrastive learning with Multiple Negatives Ranking Loss, which pulls a query representation closer to its positive passage while pushing it away from in-batch negatives. When labelled relevance judgements are scarce, synthetic query generation can provide the training signal needed for fine-tuning. This approach, which uses a generative model to produce plausible queries for each passage [Bonifacio et al., 2022], makes domain adaptation practical even when annotated data is limited.

2.1.4 Synthetic Query Generation

In the absence of annotated query sets, constructing a realistic evaluation and training data without manual annotation effort is a recurring challenge. Bonifacio et al. [2022] demonstrated that LLMs can serve as effective synthetic query generators: by conditioning a model on a document with few-shot examples, they produced query-document pairs that, when used for fine-tuning, outperformed BM25 and self-supervised dense retrieval baselines.

However, single-prompt generation approaches tend to produce queries that cluster around surface-level document features, limiting lexical and intent diversity. Ge et al. [2024] address this by conditioning generation on synthetic personas rather than documents alone, explicitly varying searcher background, goal, and vocabulary to broaden the coverage of the resulting query set. Query diversity is not

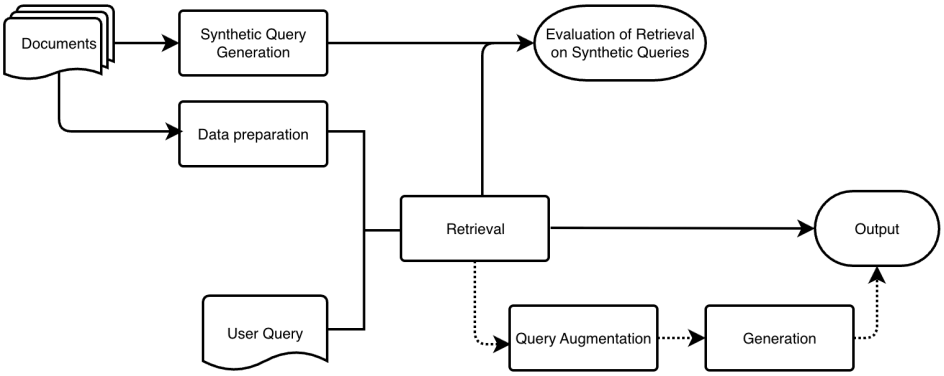


Figure 2.1: Visualisation of the whole rag pipeline.

guaranteed by scale alone, structured variation in the generation process through persona conditioning helps produce evaluation sets representative of real retrieval scenarios.

2.2 Retrieval

The retrieval stage is the core component of a RAG pipeline, responsible for identifying and returning the most relevant documents from a knowledge base given a user query. Retrieval approaches broadly fall into three categories: sparse, dense, and hybrid methods. Sparse retrievers such as **BM25** [Robertson and Zaragoza, 2009] rely on lexical overlap between query and document tokens, offering strong performance on keyword-heavy queries with minimal computational overhead. Dense retrievers, by contrast, encode both queries and documents into a shared vector space using neural encoders, enabling semantic matching that is robust to lexical mismatch [Karpukhin et al., 2020]. Bi-encoder architectures are the dominant paradigm for dense retrieval: a query encoder and a document encoder produce independent representations, and relevance is computed via a lightweight similarity function such as cosine similarity or dot product, making the approach practical for large-scale retrieval [Lewis et al., 2020].

Among dense encoders, **E5** [Wang et al., 2022] has emerged as a strong general-purpose bi-encoder trained via contrastive learning that consistently ranks among the top-tier models on the Massive Text Embedding Benchmark (MTEB) [Muennighoff et al., 2023], a standardised suite that enables reliable cross-model comparisons. Its utility extends beyond general web retrieval: Xu et al. [2024] demonstrate strong performance using E5 for specialised technical documentation retrieval, a setting that shares structural properties with other narrow-domain search tasks.

Information retrieval literature broadly supports the fusion of sparse and dense retrievers into hybrid systems. Such architectures, which combine the exact-matching strengths of BM25 with the semantic generalisation of dense embeddings, consistently outperform either component in isolation across diverse query types [Sharma, 2025]. Several fusion strategies have been proposed. Score fusion combines retrieval scores directly after normalisation, making it sensitive to score distribution mismatches between retrievers. Reciprocal Rank Fusion (RRF) instead aggregates ranked lists without requiring score normalisation, assigning each document a fusion score based on its rank position across retrievers [Cormack et al., 2009]; a weighted variant allows differential contribution of each retriever to the final ranking [Samuel et al., 2025]. Cross-encoder rerankers take a different approach entirely: a small candidate set from each retriever is passed jointly with the query into a fine-tuned transformer model, which scores each document through full query-document interaction. While cross-encoders generally achieve higher ranking precision, they incur substantially greater computational cost, making them less suitable for low-latency retrieval settings.

2.3 Generation

The generation component of a RAG framework is responsible for transforming retrieved information and the user query into a natural, coherent response. It acts as a decoder that synthesises an answer by combining its internal knowledge with the specific evidence provided by the retriever [Gao et al., 2023]. This grounding in retrieved context is the primary mechanism by which RAG systems mitigate hallucination, the tendency of standard LLMs to produce fluent but factually unsupported responses.

Unlike standard LLMs that take only a query as input, a RAG generator is conditioned on two sources: parametric memory, the general knowledge encoded in the model's pre-trained weights, and non-parametric memory, the external document chunks retrieved from the knowledge base [Lewis et al., 2020]. In practice, these are combined by concatenating the user query and the retrieved documents into a single prompt. The design of this prompt is a non-trivial decision, as even subtle differences in wording or structure can measurably affect response quality [Li et al., 2025].

2.4 Evaluation

Contemporary literature frequently evaluates RAG systems along three core dimensions: context relevance, answer faithfulness, and answer relevance [Gao et al., 2023, Sharma, 2025]. Context relevance measures whether the retrieved documents contain information appropriate to the query; answer faithfulness assesses whether the generated response is grounded in the retrieved context without introducing unsupported claims; and answer relevance evaluates whether the final response actually addresses the user's question [Gao et al., 2023].

Automated frameworks such as RAGAS and ARES operationalise these three dimensions jointly, using LLM-based judges to score each one automatically and reducing the need for costly human annotation. When the focus is on retrieval quality alone, classical Information Retrieval metrics such as Normalised Discounted Cumulative Gain (nDCG), Recall and Mean Reciprocal Rank (MRR) are commonly used instead. These metrics quantify both the relevance and the ranking quality of retrieved documents, and have been adopted as standard evaluation protocols in recent retrieval-focused studies [Pandey, 2026, Xu et al., 2024, Muennighoff et al., 2023].

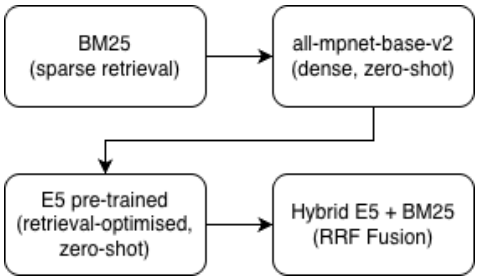


Figure 3.1: Diagram of the ablation ladder this study follows

3 Methods

This chapter describes the experimental design and implementation of the retrieval systems evaluated in this thesis. To systematically assess the effect of different retrieval strategies on equipment discovery performance, an ablation ladder is adopted, in which each condition introduces a single change over the previous one; the full ladder is illustrated in Figure 3.1. The complete implementation, including indexing scripts, evaluation pipelines, and experimental configurations, is publicly available. *

3.1 Dataset and Document corpus

The dataset used for this thesis consists of 8601 instruments, with the following features: *Equipment name, Aliases, Short description, Typical applications, Research-oriented use, Inputs, Outputs, Tags, Safety/training notes, Source links, Confidence, Confidence note, University, Authors / lab contacts*. The Table 3.1 lists the percentage of completeness for each of the categories (Visualisation can be obtained using file `/scripts/s02_visualise_data`).

Several fields exhibit near-zero completeness. Including predominantly empty fields in the retrieval pipeline risks introducing noise without informational benefit. Fields with fewer than 10% completeness were therefore excluded.

*`https://github.com/vkotuliak/bachelor_thesis_codebase`

Category	Completeness
Equipment name	100.0%
Typical applications	100.0%
Research-oriented use	100.0%
Confidence	100.0%
Tags (5-10 keywords)	99.9%
Source link(s)	99.7%
Confidence note	99.6%
Short description	97.0%
Aliases / synonyms	5.2%
University	2.1%
Authors / lab contacts	1.2%
Inputs	0.5%
Outputs	0.5%
Safety/training notes	0.5%

Table 3.1: Field completeness across the 8,601-item equipment corpus. Fields retained for retrieval are shown in bold.

The *Confidence* field encodes the estimated correctness of each scraped record. Inspecting its distribution revealed that 8,587 of the 8,601 items carry *High* confidence and 14 carry *Medium* confidence. As the 14 medium-confidence items represent less than 0.2% of the corpus and may introduce unreliable content, they were excluded, yielding a final corpus of 8,587 items. Because all retained records share the same confidence value, the field becomes uninformative for retrieval and was dropped. The related *Confidence note* field, which records the rationale behind each confidence assignment, was excluded for the same reason.

The *Source link* field stores the URL from which each record was originally scraped. As URLs carry no semantic content relevant to equipment discovery, this field was also excluded.

The fields retained for document construction were: *Equipment name*, *Short description*, *Typical applications*, *Research-oriented use*, and *Tags*. These are indicated in bold in Table 3.1.

Further work with the data included getting these categories out of the original CSV, cleaning them by removing bullet points, new lines and extra white spaces. And finally saving them into two JSON lines files, one for generation of queries intended for testing called `documents.jsonl` and second for RAG retrieval called `rag_documents.jsonl`. The file intended for generation of queries has a following format:

```
{
    "id": f"eq_{idx:04d}",
    "name": <name>,
    "short description": <description>,
    "typical applications": <applications>,
    "research-oriented use": <
    research_oriented use>,
    "tags": <tags>
}
```

The file intended for RAG retrieval has a format inspired by Pandey [2026]:

```
{"equipment_description": f"name: {name} [
    SEP] short description: {description} [
    SEP] typical applications: {
    applications} [SEP] research-oriented
    use: {research_use} [SEP] tags: {tags
    }"}
```

The difference stems from the fact that RAG retrievers retrieve from textual data and not from `json` object. On the other hand, LLMs can handle the object and it can help with more precise generation of queries. Refer to `/scripts/s01_build_documents.py` for more details.

3.2 Ablation Ladder

3.2.1 Okapi Best Matching 25 (BM25)

The ablation study begins with BM25 as the sparse retrieval baseline, the mechanics of which are described in Section 2. Prior to indexing, the corpus and queries are preprocessed through the following tokenisation pipeline:

1. Converting text to lowercase
2. Removing punctuation
3. Tokenising into individual terms
4. Removing stop-words using the `nltk` stop-word list
5. Stemming each term using `PorterStemmer` from `nltk`

This pipeline is applied uniformly to both the document corpus and all evaluation queries to ensure consistency between indexing and retrieval.

3.2.2 all-mpnet-base-v2

As the dense retrieval baseline, `all-mpnet-base-v2` is adopted, following its use in the same role by Pandey [2026]. This model serves as a natural dense baseline due to its general-purpose sentence encoding objective, which is not optimised for retrieval specifically, making it a meaningful comparison point for retrieval-oriented models. Embeddings are generated for the entire corpus and stored in a FAISS index. At retrieval time, the index is loaded, the query is embedded using the same model, and Approximate Nearest Neighbour (ANN) search is performed to return the highest-ranked matches.

3.2.3 Pre-trained E5

To evaluate whether a retrieval-optimised encoder improves upon the general-purpose dense baseline, `multilingual-e5-large` is incorporated as the next condition. Following the instructions of Wang et al. [2022], each document in the corpus is prefixed with `"passage:"` prior to embedding, and each query is prefixed with `"query:"` at retrieval time. Corpus embeddings are generated and stored in a FAISS index, and retrieval proceeds via ANN search in the same manner as the previous condition.

3.2.4 Hybrid Architecture

As discussed in Section 2, combining sparse and dense retrieval has been widely shown to improve performance over either approach in isolation. To this end, a hybrid architecture is constructed by fusing the BM25 and E5 retrievers using Weighted Reciprocal Rank Fusion (WRRF). Both retrievers independently produce a ranked list of 200 candidate documents, which are subsequently merged using the following formula:

$$WRRF(d) = \frac{\alpha}{k + r_{E5}(d)} + \frac{1 - \alpha}{k + r_{BM25}(d)}$$

where d is a document, $r_{E5}(d)$ and $r_{BM25}(d)$ are the ranks assigned to d by the E5 and BM25 retrievers respectively, k is a smoothing constant, and $\alpha \in [0, 1]$ is a weighting parameter controlling the relative contribution of each retriever [Cormack et al., 2009, Samuel et al., 2025].

WRRF was selected over score fusion and cross-encoder re-ranking for several reasons. Score fusion requires score distributions across retrievers to be compatible; given that BM25 and E5 produce scores on fundamentally different scales, normalisation introduces additional sensitivity to outliers, which is a particular concern when retrieval is performed over synthetically generated queries whose score distributions may be irregular. Cross-encoder re-ranking, while capable of higher precision, requires joint query-document inference over a candidate set, introducing latency that is undesirable in an interactive retrieval setting. WRRF avoids both issues by operating on rank positions rather than raw scores.

The weighted variant is adopted here, assigning differential weights to the BM25 and E5 ranked lists, as preliminary experiments indicated a modest but consistent improvement across all the metrics over the unweighted formulation. Weights were determined via grid search; the resulting configuration assigns a weight of 0.6 to BM25 and 0.4 to E5, with the smoothing constant k set to 10. The full grid search results are reported in Appendix B.

Each retriever returns a candidate pool of 200 documents prior to fusion. This threshold is chosen conservatively: given the smoothing constant $k = 10$, documents ranked beyond 200 positions contribute a fusion score below $\frac{1}{210} \approx 0.005$, rendering their influence on the final ranking negligible. Expanding the pool further would therefore have no meaningful effect on retrieval outcomes while increasing computational overhead.

3.3 Synthetic Query Generation

As no annotated query sets existed for the RUG equipment portal, a synthetic generation pipeline was developed to construct a representative evaluation benchmark. An initial pilot using a single prompt template with a commercial model produced 500 queries; manual inspection revealed limited lexical and intent diversity, with queries clustering around instrument names rather than researcher problems. This motivated a persona-driven approach following Ge et al. [2024], which explicitly varies searcher background and goal to broaden query coverage across specificity levels, from precise instrument requests to broader capability-based queries.

The pipeline consists of three steps: *text-to-persona*, *deduplication*, and *query generation*. In the *text-to-persona* step, equipment metadata is provided to `qwen2.5:7b` with instructions to generate five personas representing distinct researcher types who might search for that item; full prompt templates are provided in Appendix A. Five personas per item were chosen to match the target of five queries per equipment, ensuring each generated query reflects a distinct searcher perspective.

Prior to query generation, a *deduplication* step filters personas that are too similar to others generated for the same equipment item. Similarity is computed using `SequenceMatcher` from Python's `difflib` library with a threshold of 0.8. Given that at most five personas are compared per item, this approach is computationally appropriate and avoids the overhead of approximate hashing methods such as MinHash.

In the final *query generation* step, each remaining persona is combined with the corresponding equipment metadata to produce a single natural language query. The prompt instructs the model to adopt the persona's perspective and goal, and to avoid direct reference to equipment names, so that queries reflect how a researcher would express a problem rather than a solution.

The resulting benchmark consists of 2489 queries. As each query is generated with direct access to the metadata of its corresponding document, this construction yields a controlled synthetic benchmark rather than a sample of independently formulated, naturalistic queries. The implications of this design for retrieval performance are examined in Section 4.2.

3.4 Evaluation

Retrieval performance was evaluated using three complementary metrics: **nDCG@k**, **Recall@k**, and **MRR**, computed at $k \in \{1, 5, 10\}$. The system is intended for domain experts who, when presented with a ranked list of instruments, are likely to scan results from top to bottom and give more attention to higher-ranked entries. For this reason, nDCG@10 was selected as the primary metric: it rewards systems that place the most relevant instrument near the top of the result window, reflecting the practical advantage of surfacing the best match early. At the same time, Recall@10 was used as an important secondary metric, since failing to retrieve the relevant instrument within the top ten results entirely represents a more fundamental failure than returning it at a suboptimal rank. MRR complements both metrics by capturing the rank of the single correct instrument, which is a natural fit given that each query in the evaluation set has exactly one ground-truth document.

Ground truth for evaluation was constructed from the synthetic query generation pipeline described in Section 3.3: each query was derived from a specific instrument entry, establishing a one-to-one mapping between query and relevant document. To compute evaluation scores, each retrieval system was queried with every synthetic query in the evaluation set. The system returned a ranked list of instruments from the full dataset, and the retrieved ranking was compared against the ground-truth document associated with that query. nDCG@k, Recall@k, and MRR were then computed for each query-ranking pair and averaged across the full evaluation set to produce the final scores reported in Section 4.

4 Results & Discussion

The retrieval systems were evaluated on 2489 synthetic query–answer pairs over 8587 instruments from the collected dataset. The ablation study proceeds from BM25, a sparse retriever, through `all-mpnet`, a general-purpose dense encoder not optimised for retrieval, to E5, a retrieval-optimised dense encoder, and concludes with a hybrid architecture combining BM25 and E5 via weighted Reciprocal Rank Fusion (WRRF).

4.1 Retrieval Performance across Systems

Results presented in Table 4.1 demonstrate strong performance of BM25, which outperforms both dense encoders across all metrics, most notably `all-mpnet` by 0.18 nDCG@10 points and E5 by 0.08 nDCG@10 points. Among the dense encoders, the retrieval-optimised E5 outperforms `all-mpnet` by 0.10 nDCG@10 points, consistent with its retrieval-specific training objective.

The strong performance of BM25, which relies solely on lexical overlap between queries and instru-

System	nDCG@5	nDCG@10	R@1	R@5	R@10	MRR
BM25	0.6405	0.6564	0.5271	0.7356	0.7843	0.6216
all-mpnet-base-v2	0.4504	0.4761	0.3331	0.5520	0.6308	0.4272
E5	0.5499	0.5716	0.4359	0.6481	0.7155	0.5260
Hybrid (E5 + BM25)	**0.6646**	**0.6845**	**0.5396**	**0.7702**	**0.8313**	**0.6431**

Table 4.1: Retrieval performance across systems

ment descriptions, suggests that despite the effort invested in generating varied synthetic queries, the generated queries retain substantial lexical similarity to the source descriptions. Manual inspection of the query set supports this, revealing direct lexical matches, such as domain-specific terms like *ultrastructure* or *surface topography*, appearing verbatim in both query and description. The extent and implications of this overlap are examined quantitatively in Section 4.2.

The underperformance of dense retrievers relative to BM25 can be partly attributed to the fact that neither `all-mpnet` nor E5 was fine-tuned on the equipment corpus. Both models were applied in a zero-shot setting, meaning their embedding spaces were not adapted to the domain-specific semantic relationships present in equipment descriptions. Fine-tuning on in-domain query-description pairs, particularly with hard negatives, would likely improve the performance of dense retrieval.

The hybrid architecture, combining BM25 and E5 via WRRF, achieves an nDCG@10 score of 0.6845, outperforming BM25 alone by 0.03 nDCG@10 points and E5 alone by 0.11 nDCG@10 points, and represents the best-performing configuration overall. Recall@10 and MRR rankings are consistent with nDCG@10 across all conditions, confirming the robustness of these findings.

This result is likely explained by the complementary failure modes of the two retrievers: when BM25 misses a relevant instrument due to a lack of exact lexical overlap, E5 can recover it through semantic similarity, and vice versa. Because these failure modes rarely coincide, their combination covers a broader range of queries than either retriever alone.

To understand why the hybrid architecture performs the best, a closer inspection is facilitated by a grid search over k and α (Table B), the two parameters governing WRRF behaviour. The optimal value of k=10, substantially lower than the conventional k=60, indicates that WRRF heavily rewards documents ranked highly by at least one retriever. For instance, an instrument ranked 2nd by E5 but 150th by BM25 receives a higher fusion score than one ranked 20th by both retrievers (0.09 vs. 0.066, assuming equal weights; it would be opposite with conventional k=60). Low k functions less as an averaging mechanism and more as a rescue mechanism: it allows a strong signal from one retriever to dominate the fused score even when the other retriever ranks the document poorly.

Turning to α, the best configurations cluster around a moderate BM25 weighting ($\alpha_{\text{E5}} = 0.4$, $\alpha_{\text{BM25}} = 0.6$), while E5-heavy configurations ($\alpha_{\text{E5}} \geq 0.6$) perform consistently worse across all values of k, suggesting that BM25's lexical precision plays an indispensable role that semantic similarity alone cannot replicate. Together, the optimal k and α values indicate that the hybrid system performs best when BM25 contributes lexical precision and low-k WRRF ensures that strong signals from either retriever are not diluted by the other's failures.

These results suggest that the hybrid architecture is the most effective configuration for this corpus, driven by the complementary failure modes of BM25 and E5 and the error-dampening effect of low-k WRRF.

4.2 Query–Document Leakage and Implications for Retrieval

To quantify the lexical overlap suggested by manual inspection, stem-level overlap between each synthetic query and its corresponding instrument description was computed across the full evaluation set. On average, 50.88% of the stemmed terms in a query also appear in the corresponding description, indicating that queries share roughly half of their vocabulary with the document they are meant to retrieve.

This magnitude is consistent with two properties of the generation pipeline. First, while the query generation prompt explicitly restricted the use of instrument names, no equivalent restriction was placed on domain-specific terminology, allowing scientific vocabulary to persist directly from description to query. Second, following the PersonaHub methodology, personas were generated to reflect realistic researcher backgrounds; expert personas in particular retained technical jargon, and since queries were generated conditioned on these personas, this jargon was inherited into the final queries, further increasing lexical overlap with the source descriptions.

This overlap has direct implications for interpreting the ablation results. BM25 retrieves purely via lexical matching, so any shared vocabulary between a query and its target description directly and disproportionately benefits it; the nDCG@10 gap observed between BM25 and both dense encoders should therefore be read partly as a property of this benchmark's construction, rather than as a general finding about lexical versus semantic retrieval in the equipment domain.

5 Conclusions

Finding relevant equipment in the University of Groningen's growing equipment database using keyword-based search is insufficient, particularly for researchers without prior knowledge of available instrumentation. This work investigated how a Retrieval-Augmented Generation pipeline can address this limitation by evaluating four retrieval strategies across a systematic ablation study.

To assess each strategy in isolation, an ablation ladder was constructed, progressing from the sparse retriever BM25, through a general-purpose dense encoder `all-mpnet` and a retrieval-oriented dense encoder E5, to a hybrid architecture fusing BM25 and E5 via weighted reciprocal rank fusion. Within this benchmark, the hybrid system consistently achieved the strongest performance across all evaluation metrics, demonstrating that combining sparse and dense retrieval improves equipment discovery accuracy over either retriever alone under the conditions evaluated. BM25 also outperformed both dense encoders in isolation; however, as discussed in Section 4.2, this advantage is partly attributable to lexical overlap introduced by the synthetic query generation process, and should not be read as a general claim about retrieval settings involving domain-specific terminology.

These results indicate that the proposed pipeline is a promising foundation for equipment discovery within the assumptions of this controlled, synthetic benchmark. Whether these improvements translate to real-world researcher queries, and thus to practical deployment, remains to be validated and is left as an important direction for future work.

6 Limitations

6.1 Synthetic dataset

The main limitation of this study concerns the evaluation dataset. No human-annotated gold labels were available for this corpus, so synthetic query-answer pairs were generated instead to serve as ground truth. While considerable effort went into constructing a faithful and varied dataset through a multi-step generation process, this approach lacks the independent validation that human relevance judgements would provide. As a result, reported performance is evaluated relative to the synthetic generation process itself, rather than against an external standard of relevance.

A related but distinct concern is whether the distribution of synthetic queries reflects how real researchers would actually search. Since queries were generated by prompting an LLM with expert personas, they likely skew toward technically articulate, well-formed phrasing. Real researchers, particularly those unfamiliar with the relevant equipment, might query with vague, incomplete, or colloquial language instead. This discrepancy is particularly relevant given that BM25 performed strongly in this evaluation, a result driven by lexical overlap between expert-phrased queries and technical instrument descriptions; this advantage may not hold to the same extent for less technically articulate, real-world queries.

6.2 Single corpus evaluation

A further limitation concerns the use of a single corpus for both development and evaluation. All conclusions regarding the relative performance of

retrieval strategies, including the superiority of the hybrid architecture, are drawn from this one institutional dataset of approximately 8,600 instrument entries. Whether these findings extend to other domain-specific or technically dense corpora remains untested.

This concern is compounded by the fact that the WRRF hyperparameters governing the hybrid architecture, α and k, were themselves tuned via grid search on this same corpus and evaluation set. Consequently, the reported configuration may be specific to this dataset's characteristics, and an optimal weighting elsewhere may differ substantially. This further amplifies the generalisability concern specifically for the hybrid architecture, since its strong performance is conditional on hyperparameters that have not been validated outside this setting.

6.3 No fine-tuning for dense encoders

This project relied exclusively on pre-trained `all-mpnet` and E5, without fine-tuning either encoder on domain-specific data. While both models performed competitively out-of-the-box, their embedding spaces were not optimised for the specialised terminology of scientific instrumentation. As such, the reported performance of dense retrieval likely represents a lower bound: fine-tuning on in-domain query-document pairs could narrow or even close the gap observed between dense and hybrid retrieval in this evaluation.

7 Future Work

7.1 Evaluate using human-created pairs

A natural next step would be to collect real query-answer pairs from researchers using the equipment portal and re-evaluate the system on this human-annotated set. This would address two distinct shortcomings of the present evaluation.

First, it would provide an external check on the validity of the synthetic relevance judgements used in this study. Since the synthetic dataset was generated and labelled by the same underlying process, performance is currently measured relative to the generation pipeline itself rather than against an independent standard of relevance. Human-annotated pairs, collected from actual portal usage and judged by domain experts, would offer this missing external validation and clarify whether the synthetic benchmark is a trustworthy proxy for real relevance.

Second, it would test whether the present findings hold under a more realistic query distribution, addressing the concern raised in Section 6.1 regarding the technical phrasing of synthetic queries. Collecting a human-annotated set would allow this assumption to be tested directly, both by evaluating retrieval performance on more realistic queries and by quantifying the distributional gap between the two query sets, for instance in terms of vocabulary overlap or lexical complexity, to characterise how the synthetic queries diverge from real-world usage. Evaluating on human-created pairs would then directly test whether the relative strength of BM25, and consequently the composition of the hybrid system's advantage, persists under these more realistic conditions. In this sense, the two motivations reinforce one another: a human-annotated set would not only validate the labels themselves but also determine whether the strategy ranking observed in this study is an artefact of the synthetic query distribution or a genuine, generalisable result.

7.2 Fine-tuning E5

A further direction is fine-tuning E5 on in-domain query-document pairs, rather than relying on its pre-trained embeddings. As noted in Section 6.3, this could narrow the performance gap between E5 and the hybrid system. The size of any remaining gap would also be informative: if fine-tuning closes it substantially, this would suggest BM25 was largely compensating for E5's lack of domain adaptation; if a gap persists, it would support the interpretation that lexical and semantic retrieval capture genuinely complementary signals.

7.3 Re-ranking on top of hybrid retrieval

Future work could also introduce a re-ranking stage on top of the hybrid retriever's output. While WRRF fuses BM25 and E5 scores efficiently

across a large candidate pool, a cross-encoder reranker could model query-document interactions more precisely over a small top-k subset, potentially improving precision at the cost of additional computation.

7.4 End-to-end RAG evaluation

Finally, this thesis evaluated retrieval in isolation; whether these gains translate into better generated answers remains untested and falls outside its scope. Future work could evaluate the full RAG pipeline using frameworks such as RAGAS or ARES, assessing metrics like answer relevance and faithfulness to determine whether retrieval improvements propagate to user-facing outcomes.

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A Appendix - Prompts for Query Generation

```
def build_persona_gen_prompt(item: dict) ->
    str:
    return f"""You are helping build a
    search query test set for a university
    scientific equipment portal.

Given the equipment below, list exactly 5
    distinct types of people who might
    search for it.
Describe each person by their *specific
    problem or goal*, not their job title.
The descriptions should be concrete and
    varied - different industries, career
    stages, and levels of technical
    knowledge.

<document>
Name: {item.get("name", "")}
Short Description: {item.get("short
    description", "")}
Typical Applications: {item.get("typical
    applications", "")}
Research-Oriented Use: {item.get("research-
    oriented use", "")}
Tags: {item.get("tags", "")}
</document>

Return ONLY a JSON array with exactly 5
    objects:
[
  {{"name": "short label for this persona
    type", "description": "their specific
    problem or goal in 1-2 sentences"}},
  ...
]"""
```

```
def build_query_prompt(item: dict, persona:
    dict) -> str:
    return f"""You are {persona['name']}
    with following problem/goal:

{persona['description']}

You have never heard of the equipment
    described below, but you have the
    problem above.
Write the search query you would actually
    type into a university equipment portal
    search box.

Rules:
1. Do NOT include the equipment name or any
    technical instrument names.
2. Use first-person, problem-driven
    language (e.g., "I need to...", "We are
    trying to...").
3. Write at the vocabulary level of the
    persona, avoid jargon unless it fits
    their background.
4. Focus on the scientific or practical
    problem, not on the solution.

<document>
Name: {item.get("name", "")}
Short Description: {item.get("short
    description", "")}
Typical Applications: {item.get("typical
    applications", "")}
Research-Oriented Use: {item.get("research-
    oriented use", "")}
Tags: {item.get("tags", "")}
</document>

Return ONLY a JSON object:
{{"query": "The actual search query string"
    }}
"""
```

B Grid Search

Hyperparameter Tuning Results for Hybrid Retrieval (E5 + BM25) on *part of the test dataset* (first 500 queries)
> Ordered by NDCG@10

α_{E5}	α_{BM25}	k	nDCG@10 ↓	MRR	Recall@10
0.4	**0.6**	**10**	**0.760136**	0.724598	0.883300
0.2	0.8	60	0.758277	0.724150	0.875252
0.4	0.6	5	0.758273	**0.727259**	0.869215
0.5	0.5	10	0.755597	0.715882	**0.891348**
0.2	0.8	30	0.754737	0.722608	0.867203
0.2	0.8	5	0.754598	0.723691	0.865191
0.4	0.6	30	0.754561	0.717300	0.883300
0.5	0.5	5	0.753915	0.719266	0.875252
0.2	0.8	10	0.751653	0.720047	0.863179
0.4	0.6	60	0.750187	0.714050	0.877264
0.5	0.5	30	0.748483	0.708892	0.885312
0.5	0.5	60	0.742478	0.706150	0.871227
0.6	0.4	30	0.740388	0.705647	0.865191
0.6	0.4	60	0.738346	0.703875	0.861167
0.6	0.4	5	0.734354	0.692647	0.879276
0.6	0.4	10	0.733639	0.697264	0.865191
0.8	0.2	60	0.704397	0.671736	0.822938
0.8	0.2	30	0.690919	0.659351	0.808853
0.8	0.2	10	0.676107	0.643994	0.800805
0.8	0.2	5	0.665914	0.625955	0.814889

Table B.1: Grid search results